Thinking about loss...

A way to measure your accuracy

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Matching X to Y

X = { -1, 0, 1, 2, 3, 4 }

Y = { -3, -1, 1, 3, 5, 7 }



Make a guess!

Y = 3X - 1

X = { -1, 0, 1, 2, 3, 4 }

Y = { -4, -1, 2, 5, 8, 11 }

How good is the guess?

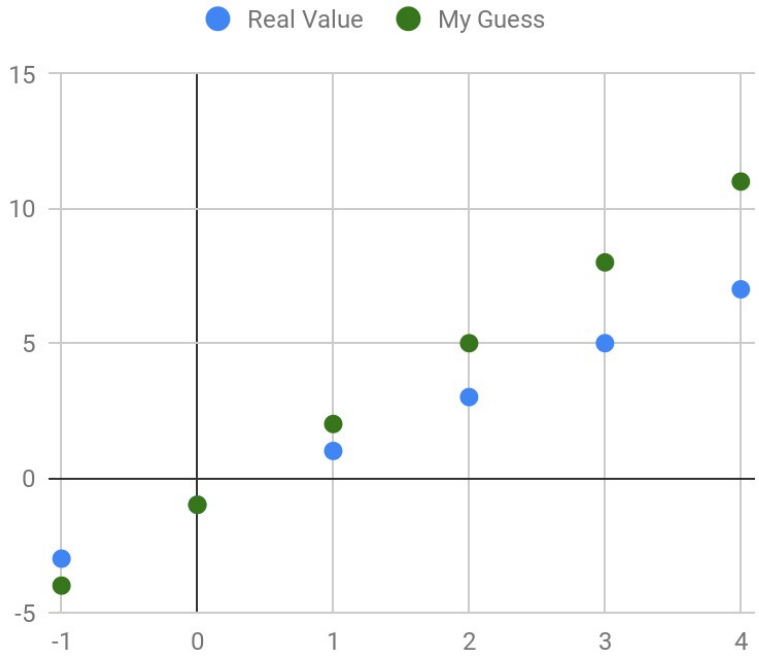
Y = 3X - 1

X = { -1, 0, 1, 2, 3, 4 }

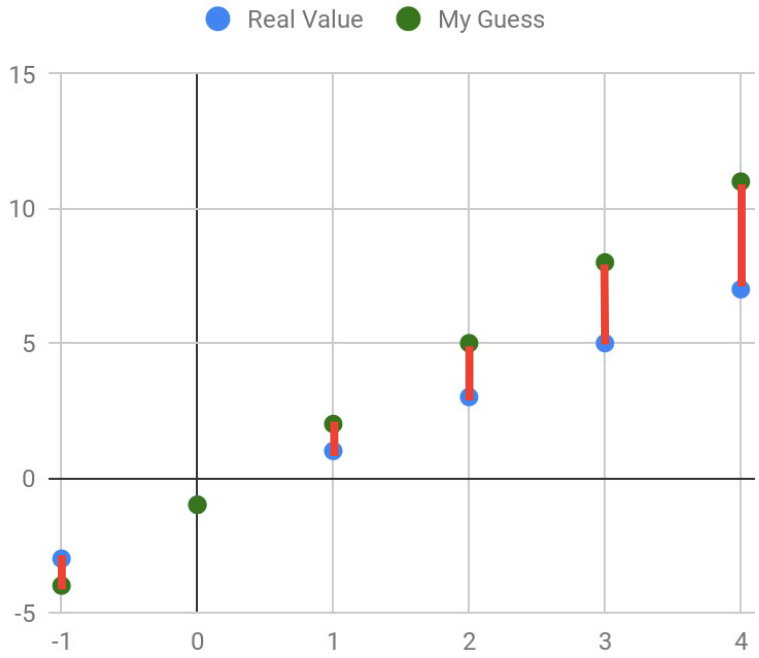
My Y = { -4, -1, 2, 5, 8, 11 }

Real Y = { -3, -1, 1, 3, 5, 7 }

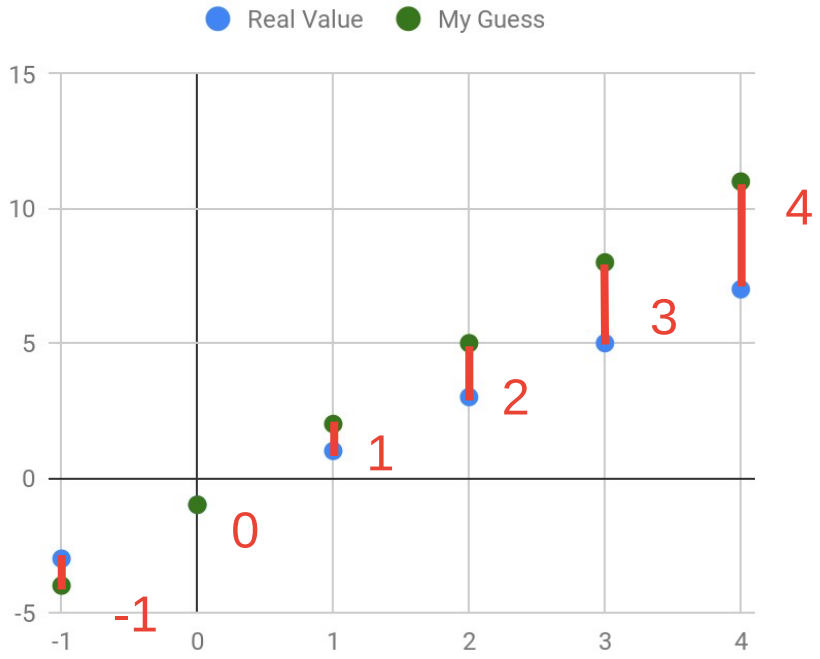
Let’s measure it!



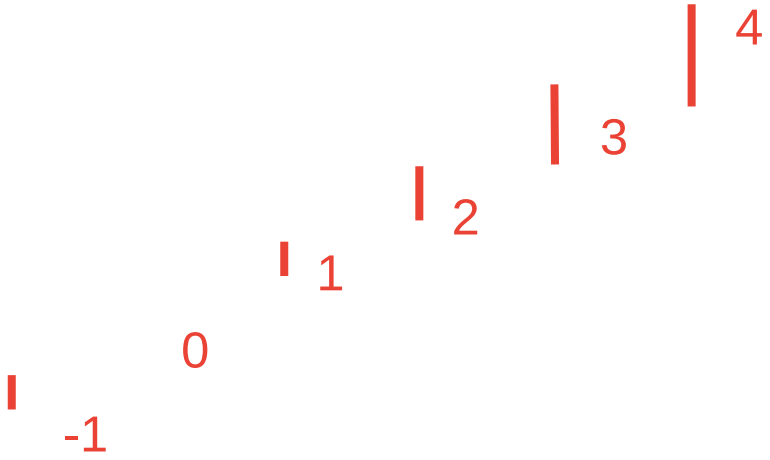
Let’s measure it!



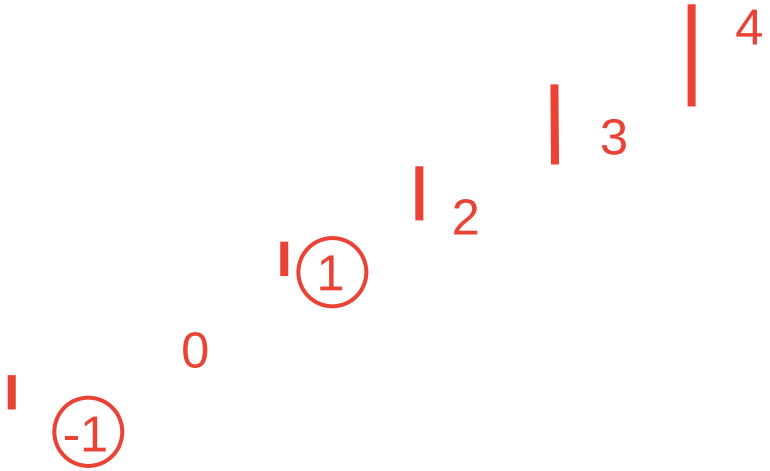
Let’s measure it!



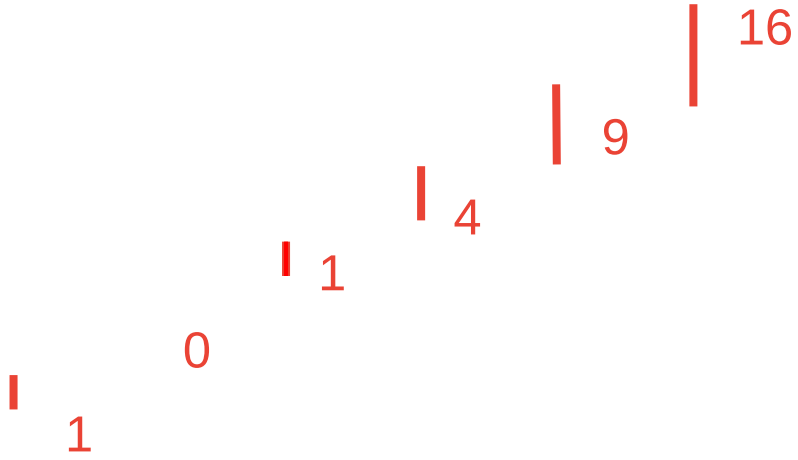
Let’s measure it!



Houston, we have a problem!



What if we **square**^{2} them?





Total that (∑) and take the square root √

sqrt(1 + 1 + 4 + 9 + 16)

= sqrt(31)
= 5.57



Make another guess!

Y = 2X - 2

X = { -1, 0, 1, 2, 3, 4 }

My Y = { -4, -2, 0, 2, 4, 6 }

Real Y = {-3, -1, 1, 3, 5, 7}

$Diff^2$ = {1, 1, 1, 1, 1}



Get the same difference, repeat the same process.

sqrt(1 + 1 + 1 + 1 + 1)

= sqrt(5)
= 2.23



Make another guess!

Y = 2X - 1

X = { -1, 0, 1, 2, 3, 4 }

My Y = { -3, -1, 1, 3, 5, 7 }

Real Y = {-3, -1, 1, 3, 5, 7}

$Diff^2$ = {0, 0, 0, 0, 0}



Your turn!